Visualization I

CONTENTS

- Visualizations in Data 8 and Data 100 (so far)
- Goals of Visualization
- An Overview of Distributions
- Variable Types Should Inform Plot Choice
- Qualitative Variables: Bar Plots
  - Plotting in Pandas
  - Plotting in Matplotlib
  - Plotting in `Seaborn`
- Distributions of Quantitative Variables
  - Histograms
    - Plotting Histograms
    - Overlaid Histograms
    - Interpreting Histograms
  - Box Plots
  - Side-by-Side Box plots
  - Violin Plots
- Kernel Density Estimation

Learning Outcomes

- Understand the theories behind effective visualizations and start to generate plots of our own with `matplotlib` and `seaborn`.
- Analyze histograms and identify the skewness, potential outliers, and the mode.
- Use `boxplot` and `violinplot` to compare two distributions.

Distributions of Quantitative Variables

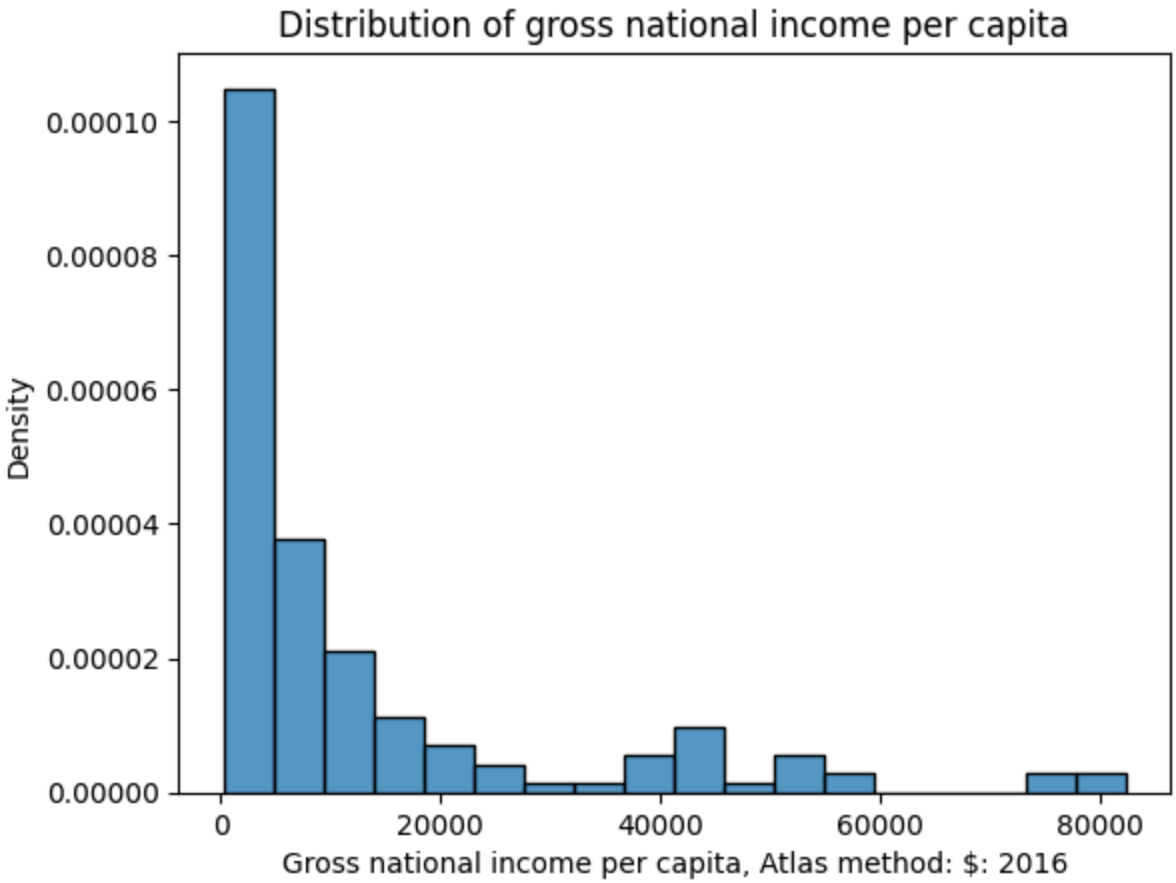
	Continent	Country	Primary completion rate: Male: % of relevant age group: 2015	Primary completion rate: Female: % of relevant age group: 2015	Lower secondary completion rate: Male: % of relevant age group: 2015	Lower secondary completion rate: Female: % of relevant age group: 2015	Youth literacy rate: Male: % of ages 15-24: 2005-14	Youth literacy rate: Female: % of ages 15-24: 2005-14	Adult literacy rate Male % ages 15 and older 2005-14
0	Africa	Algeria	106.0	105.0	68.0	85.0	96.0	92.0	83.0
1	Africa	Angola	NaN	NaN	NaN	NaN	79.0	67.0	82.0
2	Africa	Benin	83.0	73.0	50.0	37.0	55.0	31.0	41.0
3	Africa	Botswana	98.0	101.0	86.0	87.0	96.0	99.0	87.0
5	Africa	Burundi	58.0	66.0	35.0	30.0	90.0	88.0	89.0

5 rows × 47 columns

How should we define our categories for this variable? In the previous example, these were a few unique values of the `Continent` column. If we use similar logic here, our categories are the different numerical values contained in the `Gross national income per capita` column.

Under this assumption, let's plot this distribution using the `seaborn.countplot` function.

```
plot(data=wb, x="Gross national income per capita, Atlas method: $: 2016", stat="density")
e("Distribution of gross national income per capita");
```



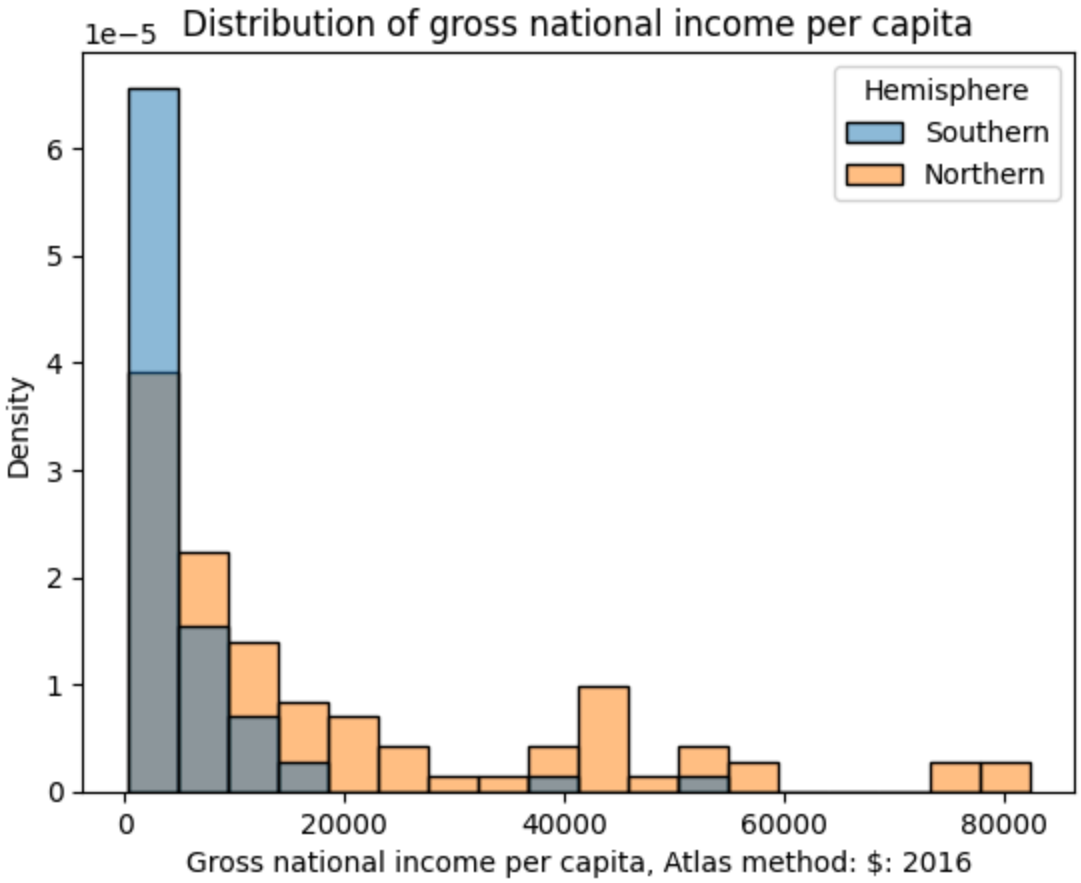
Overlaid Histograms

We can overlay histograms (or density curves) to compare **quantitative** distributions across **qualitative** categories.

The `hue` parameter of `sns.histplot` specifies the column that should be used to determine the color of each category. `hue` can be used in many `seaborn` plotting functions.

Notice that the resulting plot includes a legend describing which color corresponds to each hemisphere – a legend should always be included if color is used to encode information in a visualization!

```
sns.histplot(data=wb, x="Gross national income per capita, Atlas method: $: 2016", hue="He
plt.title("Distribution of gross national income per capita");
```

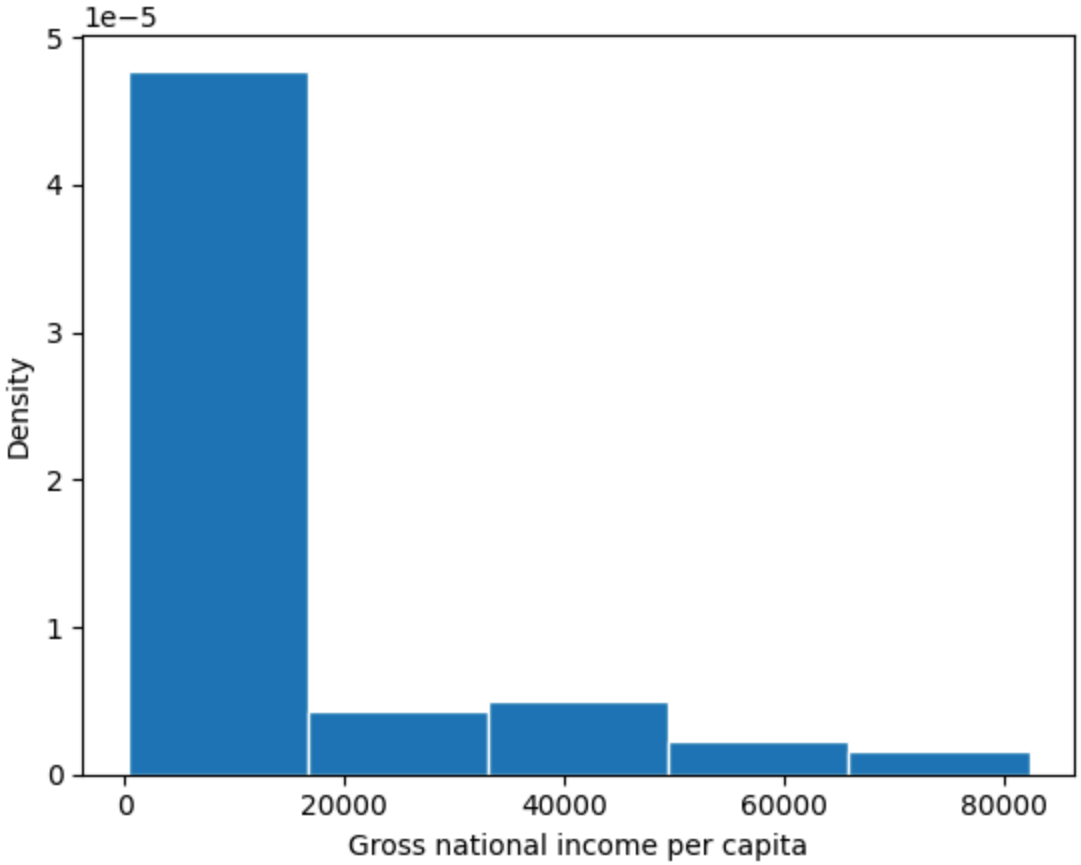


Again, each bin of a histogram is scaled such that its **area** is proportional to the **percentage** of all datapoints that it contains.

```
densities, bins, _ = plt.hist(gni, density=True, edgecolor="white", bins=5)
plt.xlabel("Gross national income per capita")
plt.ylabel("Density")
```

```
print(f"First bin has width {bins[1]-bins[0]} and height {densities[0]}")
print(f"This corresponds to {bins[1]-bins[0]} * {densities[0]} = {(bins[1]-bins[0])*densit
```

```
First bin has width 16410.0 and height 4.7741589911386953e-05
This corresponds to 16410.0 * 4.7741589911386953e-05 = 78.343949044586% of the data
```

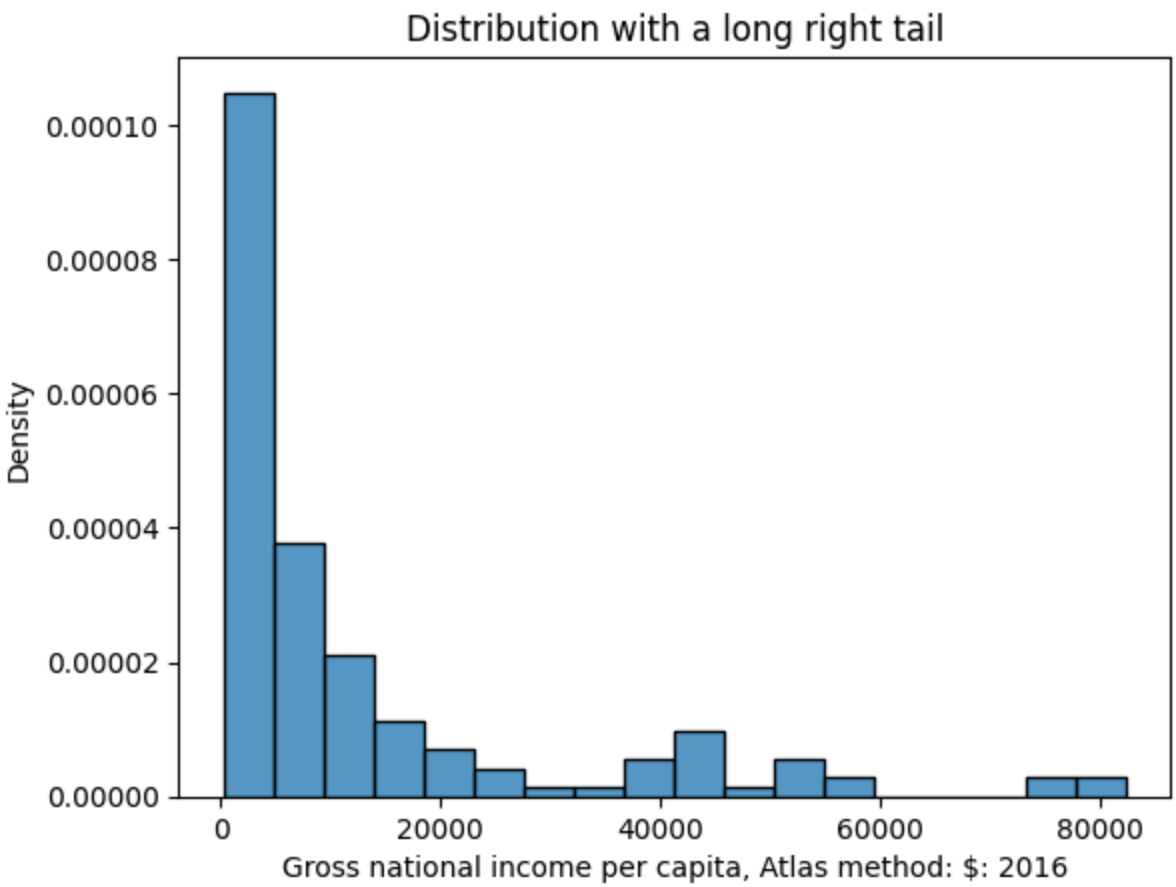


Interpreting Histograms

Histograms allow us to assess a distribution by their shape. There are a few properties of histograms we can analyze:

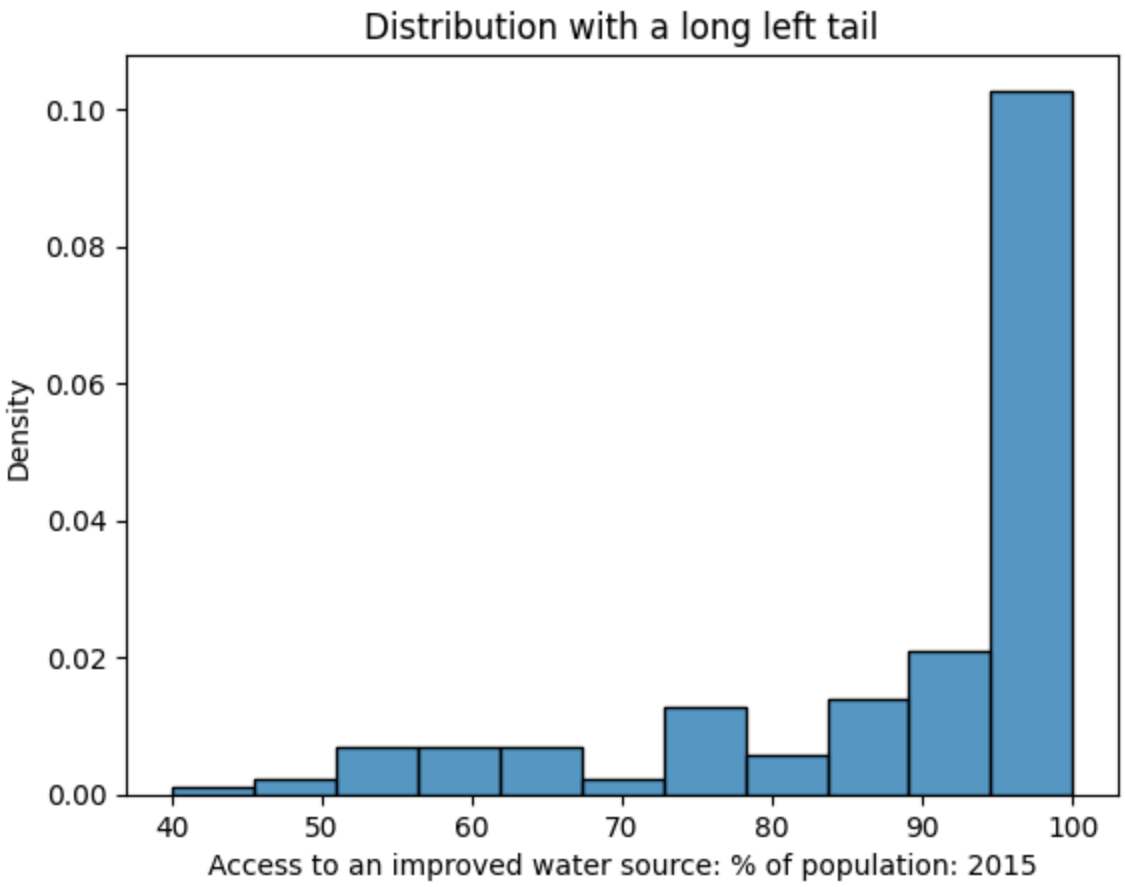
1. Skewness and Tails
    - Skewed left (negative skew) vs skewed right (positive skew)
    - Left tail vs right tail
2. Modes
    - Most commonly occurring data
3. Outliers

```
sns.histplot(data = wb, x = 'Gross national income per capita, Atlas method: $: 2016', sta
plt.title('Distribution with a long right tail');
```



- A distribution with a long left tail is **skewed left** (has **negative skew**) (such as `Access to an improved water source`). In a left-skewed distribution, the few small outliers "pull" the **mean**

```
sns.histplot(data = wb, x = 'Access to an improved water source: % of population: 2015', s
plt.title('Distribution with a long left tail');
```

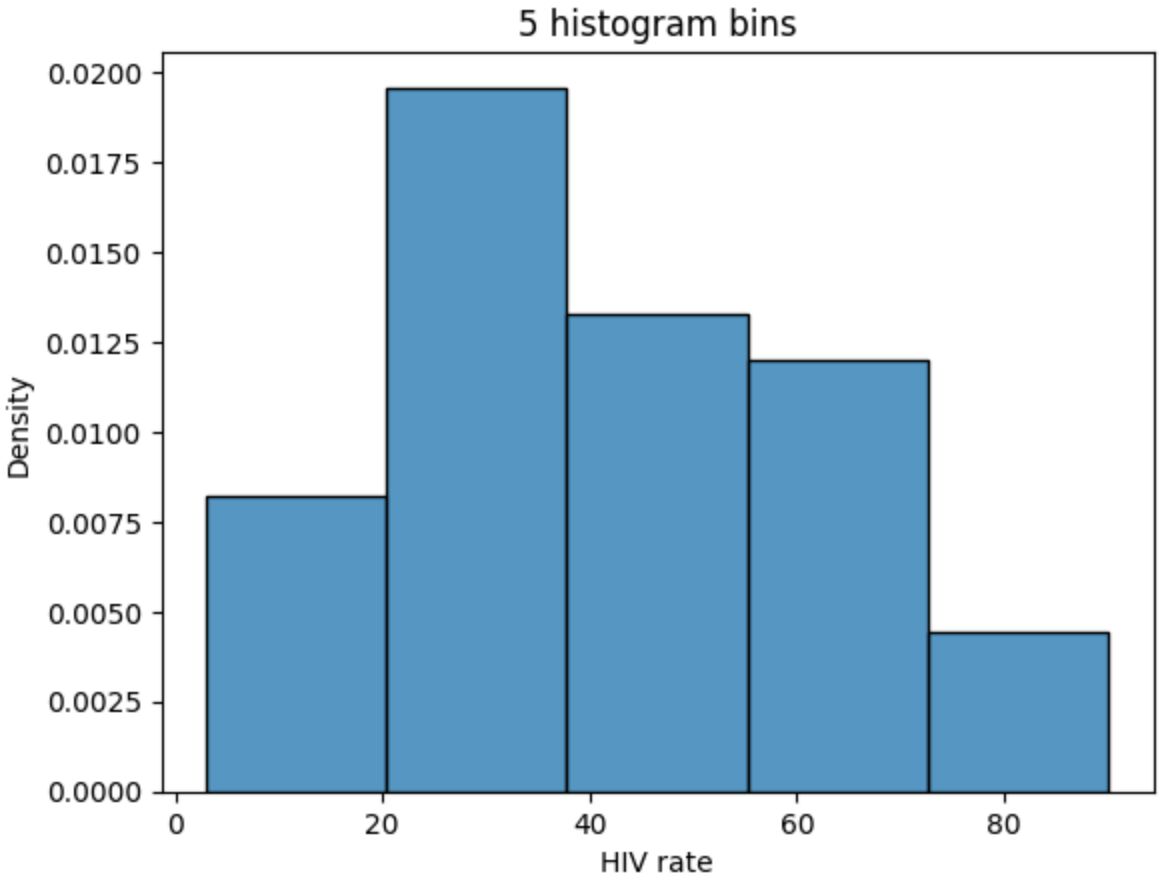


In the case where a distribution has equal-sized right and left tails, it is **symmetric**. The mean is approximately **equal** to the median. Think of mean as the balancing point of the distribution.

Modes

The **mode** is the **mo**st frequent value in a distribution. In Data 100, we describe a "mode" of a histogram as a peak in the distribution. As there may be multiple "peaks" in a distribution, we can talk about multiple modes. Often, however, it is difficult to determine what counts as its own "peak." For example, the number of peaks in the distribution of HIV rates across different countries varies depending on the number of histogram bins we plot.

```
# Rename the very long column name for convenience
wb = wb.rename(columns={'Antiretroviral therapy coverage: % of people living with HIV: 201
# With 5 bins, it seems that there is only one peak
sns.histplot(data=wb, x="HIV rate", stat="density", bins=5)
plt.title("5 histogram bins");
```




```
# With 10 bins, there seem to be two peaks

sns.histplot(data=wb, x="HIV rate", stat="density", bins=10)
plt.title("10 histogram bins");
```

Data 100: Principles and Techniques of Data Science
Regular Expressions

Data 100: Principles and Techniques of Data Science
Visualization II